

Specialist Mathematics · Medium paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ Date _____

Exam 1 ·

43 marks

reading 15 min · writing 60 min

Exam 2 ·

79 marks

reading 15 min · writing 120 min

State average on these questions

74 / 122 (61%)

a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright (some exams include third-party material).
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Methods and science working can be typed in plain text, e.g. $v = \sqrt{2gh} = 9.9 \text{ m/s}$.
3. Keep to the real time limits above, and don't look at the guide until you've finished.
4. Save with Ctrl/Cmd + S. Use a PDF app that supports forms: Adobe Acrobat Reader, Chrome, Edge, Firefox, macOS Preview or Xodo.
5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

Every question is a real VCAA exam question, chosen by ranking all the questions from the listed exams by how Victorian students actually scored on them (percentage correct for multiple choice, state average ÷ marks for written questions). This paper takes the middle band of questions.

Sources

- [2023 Exam 1 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)
- [2024 Exam 1 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)
- [2025 Exam 1 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)
- [2023 Exam 2 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)
- [2024 Exam 2 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)
- [2025 Exam 2 Specialist Mathematics exam \(PDF\)](#) · [examiners' report](#)

Questions (technology free)

Question 1 (3 marks)

Implicit differentiation of $x \arcsin(y^2) = \pi$ at $(6, \frac{1}{\sqrt{2}})$, answer in the form $-\frac{\pi\sqrt{a}}{b}$

Find $\frac{dy}{dx}$ at the point (3 marks)

Surface area when the parametric curve $x = \frac{t^2}{4} - 1$, $y = \sqrt{3}t$ is rotated about the x -axis

Find the surface area (4 marks)

Cubic with complex coefficients $3z^3 + 2iz^2 + 3z + 2i$: verify a factor, solve, plot the roots

a. Verify $3z + 2i$ is a factor (1 mark)

b. Solve $f(z) = 0$ in Cartesian form (2 marks)

c. Plot the solutions on an Argand diagram (1 mark)

Question 4 (6 marks)

2024 VCAA Exam 1 · Question 3 · [open in the official exam PDF \(p. 4\)](#) ↗

$f(x) = \frac{(x-1)^2}{(x+1)^2}$: partial-fraction form, the turning point, then a labelled sketch

a. Show $A = 1$, $B = -4$, $C = 4$ (1 mark)

b. Coordinates of the turning point (2 marks)

c. Sketch with asymptotes and intercepts labelled (3 marks)

Question 5 (5 marks)

2024 VCAA Exam 1 · Question 6 · [open in the official exam PDF \(p. 8\)](#) ↗

Three independent normal production stages: total time, variance of total cost, probability one stage is shorter than another

a. Mean and variance of the total time (1 mark)

b. Variance of the total cost (2 marks)

c. $\Pr(W_2 < W_1)$ to 2 d.p. (2 marks)

Question 6 (4 marks)

2024 VCAA Exam 1 · Question 7 · [open in the official exam PDF \(p. 10\)](#) ↗

Separable DE $x + 2y\sqrt{x^2 + 1}\frac{dy}{dx} = 0$ with $y(0) = -2$, y as a function of x

Solve the DE (4 marks)

Question 7 (4 marks)

2024 VCAA Exam 1 · Question 8 · [open in the official exam PDF \(p. 11\)](#) ↗

$x^2y^2 + xy = 2$: show $\frac{dy}{dx} = -\frac{y}{x}$, then points where the tangent slope is -1

a. Show $\frac{dy}{dx} = -\frac{y}{x}$ (2 marks)

b. Points with slope -1 (2 marks)

Question 8 (5 marks)

2025 VCAA Exam 1 · Question 3 · [open in the official exam PDF \(p. 4\)](#) ↗

Velocity $v = \frac{t}{\sqrt{t^2 + k}}$: displacement by integration, initial acceleration, then a second particle 1 m ahead after 3 s

a. Show $x = \sqrt{t^2 + k} - \sqrt{k}$ (1 mark)

b. Initial acceleration (2 marks)

c. Find k (2 marks)

Question 9 (4 marks)

2025 VCAA Exam 1 · Question 6 · [open in the official exam PDF \(p. 7\)](#) ↗

Volume when $y = \sqrt{\frac{\arctan x}{1+x^2}}$ on $[1, \sqrt{3}]$ is rotated about the x-axis

Volume in the form $\frac{a\pi^b}{c}$ (4 marks)

Question 10 (4 marks)

2025 VCAA Exam 1 · Question 7 · [open in the official exam PDF \(p. 8\)](#) ↗

Induction proof of a formula for $2^2 + 3^2 + \dots + (n + 1)^2$

Induction proof (4 marks)

Exam 2 · CAS and a bound reference allowed (79 marks, reading 15 min, writing 120 min)

Section A – Multiple-choice questions

Open each question in the official exam PDF (click the blue link), then select A–E. One mark each.

Question 1 · 1 mark

2023 VCAA Exam 2 · Section A · Question 2 · [open in the official exam PDF \(p. 2\)](#) ↗

Values of a , b , c so that $y = \frac{x^3}{ax^2+bx+c}$ has asymptotes $y = 2x + 1$ and $x = 1$

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 2 · 1 mark

2023 VCAA Exam 2 · Section A · Question 4 · [open in the official exam PDF \(p. 3\)](#) ↗

Simplify $\frac{4a}{1+z}$ for $z = -(2a + 1) + 2ai$ into polar form

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 3 · 1 mark

2023 VCAA Exam 2 · Section A · Question 9 · [open in the official exam PDF \(p. 5\)](#) ↗

Slope of a parametric path at a given time

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 4 · 1 mark

2023 VCAA Exam 2 · Section A · Question 12 · [open in the official exam PDF \(p. 6\)](#) ↗

Velocity at a given time when $a = 1 + v$, starting from rest

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 5 · 1 mark

2023 VCAA Exam 2 · Section A · Question 19 · [open in the official exam PDF \(p. 8\)](#) ↗

Probability the total of 16 normally distributed invoices exceeds a value

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 6 · 1 mark

2023 VCAA Exam 2 · Section A · Question 20 · [open in the official exam PDF \(p. 8\)](#) ↗

Population sd from a 99% confidence interval with $n = 100$

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 7 · 1 mark

2024 VCAA Exam 2 · Section A · Question 8 · [open in the official exam PDF \(p. 4\)](#) ↗

Step size h given the third Euler estimate for $dy/dx = xy^2$

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 8 · 1 mark

2024 VCAA Exam 2 · Section A · Question 11 · [open in the official exam PDF \(p. 5\)](#) ↗

Displacement from a two-segment velocity–time graph

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 9 · 1 mark2024 VCAA Exam 2 · Section A · Question 12 · [open in the official exam PDF \(p. 6\)](#) ↗Acceleration in terms of k at a given position for $x = e^{(k-1)t}$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 10 · 1 mark2024 VCAA Exam 2 · Section A · Question 17 · [open in the official exam PDF \(p. 7\)](#) ↗

Shortest distance between two parallel lines

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 11 · 1 mark2024 VCAA Exam 2 · Section A · Question 18 · [open in the official exam PDF \(p. 7\)](#) ↗

Point where a line meets a plane

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 12 · 1 mark2024 VCAA Exam 2 · Section A · Question 19 · [open in the official exam PDF \(p. 7\)](#) ↗

Definition of a type II error

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 13 · 1 mark2025 VCAA Exam 2 · Section A · Question 3 · [open in the official exam PDF \(p. 3\)](#) ↗Constants a , b , c from an oblique asymptote and y-intercept of $\frac{x^2+a}{bx+c}$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 14 · 1 mark2025 VCAA Exam 2 · Section A · Question 11 · [open in the official exam PDF \(p. 7\)](#) ↗Domain of the solution of $\frac{dy}{dx} = x^2 y^3$, $y(1) = 3$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 15 · 1 mark2025 VCAA Exam 2 · Section A · Question 13 · [open in the official exam PDF \(p. 7\)](#) ↗

Time for a ball thrown up from 50 m to land in a truck 1 m high

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 16 · 1 mark2025 VCAA Exam 2 · Section A · Question 15 · [open in the official exam PDF \(p. 8\)](#) ↗

Equation for a parameter from the angle between two planes

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 17 · 1 mark2025 VCAA Exam 2 · Section A · Question 16 · [open in the official exam PDF \(p. 8\)](#) ↗Value of n making acceleration perpendicular to velocity at a given timeYour answer: ☐ A ☐ B ☐ C ☐ D

Question 18 · 1 mark2025 VCAA Exam 2 · Section A · Question 17 · [open in the official exam PDF \(p. 9\)](#) ↗

Velocity vector by integrating acceleration, starting from rest

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 19 · 1 mark2025 VCAA Exam 2 · Section A · Question 19 · [open in the official exam PDF \(p. 9\)](#) ↗Area of the triangle cut from the axes by $x + y + z = a$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 20 · 1 mark2025 VCAA Exam 2 · Section A · Question 20 · [open in the official exam PDF \(p. 10\)](#) ↗ $\Pr(3P + 2Q - R > 25)$ as a standard normal probabilityYour answer: ☐ A ☐ B ☐ C ☐ D

Section B – Extended-response questions

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (10 marks)2023 VCAA Exam 2 · Section B · Question 3 · [open in the official exam PDF \(p. 15\)](#) ↗

Solid formed by rotating $y^2 = x - 1$: volume, curved surface area, an “efficiency ratio” (surface $\div$ volume), then a second solid with a given volume

a.i. Integral for the volume (1 mark)**a.ii.** Volume (1 mark)**b.i.** Curved surface area as $\pi \int_a^b \sqrt{Ax - B} \, dx$ (2 marks)**b.ii.** Curved surface area to 3 d.p. (1 mark)

c. Efficiency ratio to 2 d.p. (2 marks)

d. Efficiency ratio for the second solid (3 marks)

Question 2 (10 marks)

2023 VCAA Exam 2 · Section B · Question 4 · [open in the official exam PDF \(p. 17\)](#) ↗

Logistic fish populations: partial fractions, the constant D, initial stock, a prediction, maximum growth rate, a sketch, and the effect of harvesting

a. Values of A and B (1 mark)

b. Value of D (1 mark)

c. Value of n (1 mark)

d. Q when $t = 6$ (1 mark)

e.i. $\frac{d^2Q}{dt^2}$ in terms of Q (1 mark)

e.ii. Population and time of maximum growth rate (2 marks)

f. Sketch Q against t (2 marks)

g. Maximum population with harvesting (1 mark)

Question 3 (10 marks)

2024 VCAA Exam 2 · Section B · Question 5 · [open in the official exam PDF \(p. 20\)](#) ↗

Lines and planes: a vector equation, distance from a point to a line, a plane through three points, a triangle cut off by a plane

a. Vector equation of line AB (1 mark)

b. Shortest distance from L_1 to A (3 marks)

c. Cartesian equation of plane ABC (3 marks)

d.i. Axis intercepts (1 mark)

d.ii. Area of the triangle (2 marks)

Question 4 (9 marks)

2024 VCAA Exam 2 · Section B · Question 6 · [open in the official exam PDF \(p. 22\)](#) ↗

Olive-oil filling machine: one-tailed test, p-value, decision, type II error, control limits, a confidence interval and sample size

a. Hypotheses (1 mark)

b.i. p-value to 3 d.p. (1 mark)

b.ii. Should the machine be paused? (1 mark)

c. Probability of a type II error (2 marks)

d. Values a and b (1 mark)

e. 95% confidence interval (1 mark)

f. Expected number of the 40 intervals containing μ (1 mark)

g. Minimum sample size (1 mark)

Question 5 (10 marks)

2025 VCAA Exam 2 · Section B · Question 4 · [open in the official exam PDF \(p. 18\)](#) ↗

A looping planar path: start point, direction, period, a speed formula and its maximum, tracing part of the path and its length

a. Starting point (1 mark)

b. Direction arrow at (9, 0) (1 mark)

c. First return to the start (1 mark)

d. Show the speed formula (3 marks)

e. Maximum speed (1 mark)

f. Trace the path for $t \in [0, \pi]$ (1 mark)

g. Length of the traced path (2 marks)

Question 6 (10 marks)

2025 VCAA Exam 2 · Section B · Question 5 · [open in the official exam PDF \(p. 20\)](#) ↗

Three planes: their common point, a line of intersection, distance from a point to a plane, and a family of parallel planes at a given distance

a. Point of intersection (1 mark)

b.i. Direction of the line of intersection (2 marks)

b.ii. Parametric equations (1 mark)

c. Distance from $(1, 1, 2)$ to Π_3 (2 marks)

d.i. Show Π_1 is parallel to each plane in the family (1 mark)

d.ii. Values of m (3 marks)

END OF PAPER

Marking guide follows

Stop here until you have finished the paper under exam conditions.

Marking guide · Exam 1

Questions (technology free)

Question 1 (3 marks) · state average 1.8 / 3 (60%)

2023 VCAA Exam 1 · Question 4

Find $\frac{dy}{dx}$ at the point

☐ Product and chain rule: $\arcsin(y^2) + \frac{2xy}{\sqrt{1-y^4}} \frac{dy}{dx} = 0$

☐ Substitutes: $\frac{\pi}{6} + 4\sqrt{6} \frac{dy}{dx} = 0$

☐ $\frac{dy}{dx} = -\frac{\pi\sqrt{6}}{144}$

Answer / notes: $-\frac{\pi\sqrt{6}}{144}$ ($a = 6$, $b = 144$).

State average 1.8 / 3 (60%); 37% scored full marks. Mark distribution: 0: 25% · 1: 6% · 2: 32% · 3: 37%

Your mark: _____ / 3

Question 2 (4 marks) · state average 2.2 / 4 (55%)

2023 VCAA Exam 1 · Question 7

Find the surface area

☐ $S = 2\pi \int_0^2 y \sqrt{\dot{x}^2 + \dot{y}^2} dt$ with $\sqrt{\frac{t^2}{4} + 3}$

☐ $= \sqrt{3}\pi \int_0^2 t \sqrt{t^2 + 12} dt$

☐ $= \frac{\sqrt{3}\pi}{3} [(t^2 + 12)^{3/2}]_0^2$

☐ $\pi \left(\frac{64\sqrt{3}}{3} - 24 \right)$

Answer / notes: $\pi \left(\frac{64\sqrt{3}}{3} - 24 \right)$.

State average 2.2 / 4 (55%); 31% scored full marks. Mark distribution: 0: 17% · 1: 24% · 2: 13% · 3: 15% · 4: 31%

Your mark: _____ / 4

Question 3 (4 marks) · state average 2.5 / 4 (62%)

2024 VCAA Exam 1 · Question 1

a. Verify $3z + 2i$ is a factor

☐ Shows $f\left(-\frac{2i}{3}\right) = 0$ or factorises $f(z) = (3z + 2i)(z^2 + 1)$

Answer / notes: Grouping: $z^2(3z + 2i) + (3z + 2i) = (3z + 2i)(z^2 + 1)$.

State average 0.8 / 1 (80%); 73% scored full marks. Mark distribution: 0: 27% · 1: 73%

Your mark: _____ / 1

b. Solve $f(z) = 0$ in Cartesian form

☐ $(3z + 2i)(z^2 + 1) = 0$

☐ $z = -\frac{2}{3}i, \pm i$

Answer / notes: $z = i, -i, -\frac{2}{3}i$. Show that you are solving an equation, not just factorising.

State average 1.2 / 2 (60%); 48% scored full marks. Mark distribution: 0: 30% · 1: 22% · 2: 48%

Your mark: _____ / 2

c. Plot the solutions on an Argand diagram

☐ Three points on the imaginary axis at 1, -1 and $-\frac{2}{3}i$

Answer / notes: All three lie on the imaginary axis. Read the circle spacing carefully to place $-\frac{2}{3}i$.

State average 0.5 / 1 (50%); 49% scored full marks. Mark distribution: 0: 51% · 1: 49%

Your mark: _____ / 1

Question 4 (6 marks) · state average 3.3 / 6 (55%)

2024 VCAA Exam 1 · Question 3

a. Show $A = 1, B = -4, C = 4$

☐ Writes $(x - 1)^2 = (x + 1)^2 - 4(x + 1) + 4$ (or equivalent) and divides

Answer / notes: $f(x) = 1 - \frac{4}{x+1} + \frac{4}{(x+1)^2}$.

State average 0.7 / 1 (70%); 63% scored full marks. Mark distribution: 0: 37% · 1: 63%

Your mark: _____ / 1

b. Coordinates of the turning point

☐ $f'(x) = \frac{4}{(x+1)^2} - \frac{8}{(x+1)^3} = \frac{4(x-1)}{(x+1)^3}$

☐ (1, 0)

Answer / notes: (1, 0). Differentiating the part (a) form avoids the quotient rule.

State average 1.6 / 2 (80%); 65% scored full marks. Mark distribution: 0: 12% · 1: 23% · 2: 65%

Your mark: _____ / 2

c. Sketch with asymptotes and intercepts labelled

☐ Asymptotes $x = -1$ and $y = 1$ labelled

☐ Intercepts (0, 1) and (1, 0) labelled

☐ Correct shape: both branches above the x-axis, left branch above $y = 1$, minimum touching at (1, 0)

Answer / notes: Asymptotes $x = -1, y = 1$; intercepts (0, 1), (1, 0). This sketch was done poorly: extra asymptotes and unlabelled features were common.

State average 1.0 / 3 (33%); 10% scored full marks. Mark distribution: 0: 51% · 1: 13% · 2: 25% · 3: 10%

Your mark: _____ / 3

Question 5 (5 marks) · state average 2.7 / 5 (54%)

2024 VCAA Exam 1 · Question 6

a. Mean and variance of the total time

☐ Mean 4.5 h, variance 0.5 h²

Answer / notes: Mean $1 + 1.5 + 2 = 4.5$; variance $0.09 + 0.16 + 0.25 = 0.5$.

State average 0.8 / 1 (80%); 79% scored full marks. Mark distribution: 0: 21% · 1: 79%

Your mark: _____ / 1

b. Variance of the total cost

☐ $\text{Var}(10W_1 + 20W_2 + 15W_3) = 100(0.09) + 400(0.16) + 225(0.25)$

☐ 129.25

Answer / notes: 129.25 (\$²). Square the cost coefficients *and* use variances.

State average 0.9 / 2 (45%); 37% scored full marks. Mark distribution: 0: 48% · 1: 15% · 2: 37%

Your mark: _____ / 2

c. $\Pr(W_2 < W_1)$ to 2 d.p.

☐ $W_2 - W_1 \sim N(0.5, 0.5^2)$

☐ $\Pr(Z < -1) = 0.16$

Answer / notes: $\Pr(W_2 - W_1 < 0) = \Pr(Z < -1) = \frac{1-0.68}{2} = 0.16$.

State average 1.0 / 2 (50%); 45% scored full marks. Mark distribution: 0: 45% · 1: 10% · 2: 45%

Your mark: _____ / 2

Question 6 (4 marks) · state average 2.3 / 4 (57%)

2024 VCAA Exam 1 · Question 7

Solve the DE

- ☐ Separates: $\int 2y \, dy = -\int \frac{x}{\sqrt{x^2+1}} \, dx$
- ☐ $y^2 = -\sqrt{x^2+1} + c$
- ☐ $c = 5$
- ☐ $y = -\sqrt{5 - \sqrt{x^2+1}}$ (negative root since $y(0) < 0$)

Answer / notes: $y = -\sqrt{5 - \sqrt{x^2+1}}$. Students lost the last mark by leaving $y^2 = \dots$ or picking the positive root.

State average 2.3 / 4 (57%); 27% scored full marks. Mark distribution: 0: 17% · 1: 13% · 2: 21% · 3: 22% · 4: 27%

Your mark: _____ / 4

Question 7 (4 marks) · state average 2.4 / 4 (60%)

2024 VCAA Exam 1 · Question 8

a. Show $\frac{dy}{dx} = -\frac{y}{x}$

- ☐ $2xy^2 + 2x^2y\frac{dy}{dx} + y + x\frac{dy}{dx} = 0$
- ☐ Factorises: $\frac{dy}{dx} = -\frac{y(2xy+1)}{x(2xy+1)} = -\frac{y}{x}$

Answer / notes: The $(2xy+1)$ factors cancel; that factorisation is the evidence markers wanted.

State average 1.4 / 2 (70%); 54% scored full marks. Mark distribution: 0: 15% · 1: 31% · 2: 54%

Your mark: _____ / 2

b. Points with slope -1

- ☐ $y = x$ substituted: $x^4 + x^2 - 2 = 0$
- ☐ $(1, 1)$ and $(-1, -1)$

Answer / notes: $(x^2+2)(x^2-1) = 0$, so $(1, 1)$ and $(-1, -1)$. Remember the points must lie on the curve.

State average 1.0 / 2 (50%); 32% scored full marks. Mark distribution: 0: 37% · 1: 30% · 2: 32%

Your mark: _____ / 2

Question 8 (5 marks) · state average 2.9 / 5 (58%)

2025 VCAA Exam 1 · Question 3

a. Show $x = \sqrt{t^2+k} - \sqrt{k}$

- ☐ Integrates to $\sqrt{t^2+k} + c$ and uses $x(0) = 0$ to get $c = -\sqrt{k}$

Answer / notes: Apply the initial condition explicitly: that's where the mark is.

State average 0.7 / 1 (70%); 71% scored full marks. Mark distribution: 0: 29% · 1: 71%

Your mark: _____ / 1

b. Initial acceleration

- ☐ $a = \frac{dv}{dt} = \frac{k}{(t^2+k)^{3/2}}$
- ☐ $a(0) = \frac{1}{\sqrt{k}}$

Answer / notes: $\frac{1}{\sqrt{k}}$ m/s². A wrong derivative lost marks even with the right final value.

State average 1.4 / 2 (70%); 57% scored full marks. Mark distribution: 0: 19% · 1: 24% · 2: 57%

Your mark: _____ / 2

c. Find k

- ☐ $3 - (\sqrt{9+k} - \sqrt{k}) = 1$
- ☐ $k = \frac{25}{16}$

Answer / notes: $\sqrt{9+k} = 2 + \sqrt{k} \Rightarrow \sqrt{k} = \frac{5}{4}$, so $k = \frac{25}{16}$.

State average 0.8 / 2 (40%); 28% scored full marks. Mark distribution: 0: 49% · 1: 23% · 2: 28%

Your mark: _____ / 2

Question 9 (4 marks) · state average 2.5 / 4 (62%)

2025 VCAA Exam 1 · Question 6

Volume in the form $a\pi^b / c$

☐ $V = \pi \int_1^{\sqrt{3}} \frac{\tan^{-1} x}{1+x^2} dx$

☐ Substitution $u = \tan^{-1} x$ (or parts): $\pi \left[\frac{(\tan^{-1} x)^2}{2} \right]_1^{\sqrt{3}}$

☐ $\frac{\pi}{2} \left(\frac{\pi^2}{9} - \frac{\pi^2}{16} \right)$

☐ $\frac{7\pi^3}{288}$

Answer / notes: $\frac{7\pi^3}{288}$.

State average 2.5 / 4 (62%); 38% scored full marks. Mark distribution: 0: 16% · 1: 13% · 2: 13% · 3: 20% · 4: 38%

Your mark: _____ / 4

Question 10 (4 marks) · state average 2.6 / 4 (65%)

2025 VCAA Exam 1 · Question 7

Induction proof

☐ Base case $n = 1$: both sides equal 4

☐ States the assumption for $n = k$

☐ Adds $(k+2)^2$ and simplifies to $\frac{1}{6}(k+1)(2(k+1)^2 + 9(k+1) + 13)$

☐ Conclusion for all $n \in \mathbb{N}$

Answer / notes: Both sides become $\frac{1}{6}(2k^3 + 15k^2 + 37k + 24)$. Not answered well: weak base cases and misstated assumptions were common.

State average 2.6 / 4 (65%); 29% scored full marks. Mark distribution: 0: 11% · 1: 8% · 2: 18% · 3: 34% · 4: 29%

Your mark: _____ / 4

Marking guide · Exam 2

Section A – Multiple-choice questions

✓	Q	Ans	Why	How Victoria went
☐	1	B	For large x , $y \approx \frac{x}{a} - \frac{b}{a^2}$, so $a = \frac{1}{2}$, $b = -\frac{1}{4}$. The denominator is zero at $x = 1$: $a + b + c = 0$, so $c = -\frac{1}{4}$.	59% correct A 10% · B 59% · C 13% · D 14% · E 2%
☐	2	B	$1 + \bar{z} = -2a(1 + i)$, so the expression is $\frac{-2}{1+i} = -1 + i = \sqrt{2} \operatorname{cis} \frac{3\pi}{4}$. <i>Most popular wrong answer: D (17%).</i>	62% correct A 11% · B 62% · C 6% · D 17% · E 4%
☐	3	D	$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{1/2}{2/3} = \frac{3}{4}$ at $t = 2$.	65% correct A 4% · B 13% · C 13% · D 65% · E 5%
☐	4	A	$\frac{dv}{dt} = 1 + v \Rightarrow v = e^t - 1$. At $t = \log_e(e + 1)$, $v = e$. <i>Most popular wrong answer: D (16%).</i>	54% correct A 54% · B 9% · C 12% · D 16% · E 8%
☐	5	B	Total $\sim N(12\,800, 800^2)$ (variance 16×200^2). $\Pr(T > 13\,500) = \Pr(Z > 0.875) \approx 0.191$. <i>Most popular wrong answer: C (17%).</i>	67% correct A 5% · B 67% · C 17% · D 7% · E 3%
☐	6	A	Half-width $2500 = 2.5758 \frac{\sigma}{10}$, so $\sigma \approx 9710$.	63% correct A 63% · B 12% · C 13% · D 7% · E 4%
☐	7	C	$y_1 = 1$, $y_2 = 1 + h^2$, $y_3 = y_2 + 2h^2y_2^2$. $h = 0.2$ gives $1.04 + 0.08(1.04)^2 = 1.126528$. <i>Most popular wrong answer: B (23%).</i>	60% correct A 7% · B 23% · C 60% · D 9%
☐	8	B	Signed area: $+1333\frac{1}{3}$ above the axis, $-833\frac{1}{3}$ below, net 500 m east. 1000 m adds the areas (distance, not displacement). <i>Most popular wrong answer: D (16%).</i>	58% correct A 13% · B 58% · C 12% · D 16%
☐	9	C	$a = (k - 1)^2x$, so at $x = k + 1$ it's $(k - 1)^2(k + 1) = (k^2 - 1)(k - 1)$. <i>Most popular wrong answer: B (30%).</i>	50% correct A 5% · B 30% · C 50% · D 13%
☐	10	B	Vector between points $(-3, -2, 2)$; distance $= \frac{ (-3, -2, 2) \times (1, 1, 1) }{\sqrt{3}} = \frac{\sqrt{42}}{\sqrt{3}} = \sqrt{14}$. <i>Most popular wrong answer: C (28%).</i>	58% correct A 8% · B 58% · C 28% · D 6%
☐	11	D	Substitute: $-7 + 4t = 5$, $t = 3$, giving $(-5, 4, 7)$.	68% correct A 8% · B 9% · C 14% · D 68%
☐	12	A	Type II: failing to reject H_0 when H_1 is true. (Rejecting a true H_0 is type I.) <i>Most popular wrong answer: D (15%).</i>	68% correct A 68% · B 8% · C 8% · D 15%
☐	13	A	Oblique asymptote $y = \frac{x}{b} - \frac{c}{b^2}$: $b = -2$, $c = -1$. The y-intercept $\frac{a}{c} = -2$ gives $a = 2$.	68% correct A 68% · B 8% · C 13% · D 11%
☐	14	B	$\frac{1}{2y^2} = \frac{7}{18} - \frac{x^3}{3}$ must stay positive: $x < (\frac{7}{6})^{1/3}$ (strict). <i>Most popular wrong answer: C (23%).</i>	55% correct A 12% · B 55% · C 23% · D 10%
☐	15	B	Up positive: $-49 = 20t - 4.9t^2$, $t \approx 5.80$ s. Using 50 m gives 5.83. <i>Most popular wrong answer: C (25%).</i>	55% correct A 12% · B 55% · C 25% · D 7%

✓	Q	Ans	Why	How Victoria went
☐	16	A	Angle between normals: $\frac{2a+4}{3\sqrt{a^2+16}} = \frac{2}{3}$, which simplifies to $a+2 = \sqrt{a^2+16}$. <i>Most popular wrong answer: B (20%).</i>	52% correct A 52% · B 20% · C 14% · D 14%
☐	17	C	$\vec{v} \cdot \vec{a} = -8n^2e^{-4t} + 4t$. At $t = \frac{1}{2}$: $n^2 = \frac{e^2}{4}$, $n = \frac{e}{2}$. <i>Most popular wrong answer: D (15%).</i>	67% correct A 5% · B 12% · C 67% · D 15%
☐	18	C	Include a constant vector so $\vec{v}(0) = 0$: $2 \sin 2t \vec{i} - 5(\cos 2t - 1)\vec{j} + 3(e^{-2t} - 1)\vec{k}$. <i>Most popular wrong answer: A (24%).</i>	52% correct A 24% · B 17% · C 52% · D 7%
☐	19	B	Intercepts at distance a on each axis form an equilateral triangle of side $a\sqrt{2}$: area $\frac{\sqrt{3}}{4}(2a^2) = \frac{\sqrt{3}a^2}{2}$. <i>Most popular wrong answer: A (18%).</i>	59% correct A 18% · B 59% · C 14% · D 8%
☐	20	A	Mean $-6 + 6 - 5 = -5$; variance $9(4) + 4(9) + 36 = 108$. $\Pr\left(Z > \frac{30}{6\sqrt{3}}\right) = \Pr\left(Z > \frac{5\sqrt{3}}{3}\right)$. <i>Most popular wrong answer: C (27%).</i>	54% correct A 54% · B 9% · C 27% · D 10%

Multiple-choice score: / 20

Section B – Extended-response questions

Question 1 (10 marks) · state average 6.4 / 10 (64%)

2023 VCAA Exam 2 · Section B · Question 3

a.i. Integral for the volume

☐ $\pi \int_2^5 (x-1) dx$

Answer / notes: $V = \pi \int_2^5 (x-1) dx$.

State average 0.9 / 1 (90%); 89% scored full marks. Mark distribution: 0: 11% · 1: 89%

Your mark: / 1

a.ii. Volume

☐ $\frac{15\pi}{2}$

Answer / notes: $\frac{15\pi}{2}$ (keep the π).

State average 0.9 / 1 (90%); 85% scored full marks. Mark distribution: 0: 15% · 1: 85%

Your mark: / 1

b.i. Curved surface area as $\pi \int_a^b \sqrt{Ax-B} dx$

☐ $2\pi \int_2^5 \sqrt{x-1} \sqrt{1 + \frac{1}{4(x-1)}} dx$

☐ $= \pi \int_2^5 \sqrt{4x-3} dx$

Answer / notes: $a = 2$, $b = 5$, $A = 4$, $B = 3$.

State average 1.5 / 2 (75%); 62% scored full marks. Mark distribution: 0: 13% · 1: 25% · 2: 62%

Your mark: / 2

b.ii. Curved surface area to 3 d.p.

☐ 30.846

Answer / notes: ≈ 30.846 . Set CAS to show enough decimals.

State average 0.6 / 1 (60%); 55% scored full marks. Mark distribution: 0: 45% · 1: 55%

Your mark: / 1

c. Efficiency ratio to 2 d.p.

☐ Total surface = curved area $+\pi(1)^2 + \pi(2)^2$

☐ $\frac{30.846+5\pi}{7.5\pi} \approx 1.98$

Answer / notes: ≈ 1.98 . Many forgot one or both ends, or used the wrong radii.

State average 1.0 / 2 (50%); 38% scored full marks. Mark distribution: 0: 38% · 1: 24% · 2: 38%

Your mark: _____ / 2

d. Efficiency ratio for the second solid

☐ $\pi \int_2^k (x-1) dx = 24\pi \Rightarrow k = 8$

☐ Surface $= \pi \int_2^8 \sqrt{4x-3} dx + \pi + 7\pi$

☐ Ratio ≈ 1.34

Answer / notes: $k = 8$, ratio ≈ 1.34 . The end radii are 1 and $\sqrt{7}$.

State average 1.5 / 3 (50%); 24% scored full marks. Mark distribution: 0: 27% · 1: 27% · 2: 23% · 3: 24%

Your mark: _____ / 3

Question 2 (10 marks) · state average 5.8 / 10 (58%)

2023 VCAA Exam 2 · Section B · Question 4

a. Values of A and B

☐ $A = 1, B = \frac{1}{1000}$

Answer / notes: $\frac{1}{P(1-P/1000)} = \frac{1}{P} + \frac{1/1000}{1-P/1000}$.

State average 0.5 / 1 (50%); 48% scored full marks. Mark distribution: 0: 52% · 1: 48%

Your mark: _____ / 1

b. Value of D

☐ $D = 4$

Answer / notes: $200 = \frac{1000}{1+D}$.

State average 0.6 / 1 (60%); 57% scored full marks. Mark distribution: 0: 43% · 1: 57%

Your mark: _____ / 1

c. Value of n

☐ $n = 100$

Answer / notes: $Q(0) = \frac{1000}{10}$.

State average 0.8 / 1 (80%); 79% scored full marks. Mark distribution: 0: 21% · 1: 79%

Your mark: _____ / 1

d. Q when t = 6

☐ 988

Answer / notes: $\frac{1000}{1+9e^{-6.6}} \approx 988$.

State average 0.9 / 1 (90%); 85% scored full marks. Mark distribution: 0: 15% · 1: 85%

Your mark: _____ / 1

e.i. $\frac{d^2Q}{dt^2}$ in terms of Q

☐ $\frac{121}{100}Q \left(1 - \frac{Q}{1000}\right) \left(1 - \frac{Q}{500}\right)$

Answer / notes: Chain rule: $\frac{d^2Q}{dt^2} = \frac{d}{dQ} \left(\frac{dQ}{dt} \right) \cdot \frac{dQ}{dt}$. Only 20% got this.

State average 0.2 / 1 (20%); 21% scored full marks. Mark distribution: 0: 79% · 1: 21%

Your mark: _____ / 1

e.ii. Population and time of maximum growth rate

☐ $Q = 500$

☐ $t \approx 2$ years

Answer / notes: $9e^{-1.1t} = 1 \Rightarrow t = \frac{\log_e 9}{1.1} \approx 2$.

State average 1.3 / 2 (65%); 57% scored full marks. Mark distribution: 0: 32% · 1: 11% · 2: 57%

Your mark: _____ / 2

f. Sketch Q against t

- ☐ Logistic shape from $(0, 100)$
☐ Asymptote $Q = 1000$ labelled

Answer / notes: Label the intercept $(0, 100)$ and asymptote $Q = 1000$ (not $y = 1000$).

State average $1.1 / 2$ (55%); 43% scored full marks. Mark distribution: 0: 30% · 1: 28% · 2: 43%

Your mark: _____ / 2

g. Maximum population with harvesting

- ☐ 950

Answer / notes: Set $\frac{dQ}{dt} = 0$: $1.1 \left(1 - \frac{Q}{1000}\right) = 0.055$, $Q = 950$.

State average $0.4 / 1$ (40%); 40% scored full marks. Mark distribution: 0: 60% · 1: 40%

Your mark: _____ / 1

Question 3 (10 marks) · state average 6.6 / 10 (66%)

2024 VCAA Exam 2 · Section B · Question 5

a. Vector equation of line AB

- ☐ $\vec{r} = \vec{i} - 2\vec{j} + 3\vec{k} + \lambda(\vec{i} - 3\vec{j} - 4\vec{k})$

Answer / notes: Write $\vec{r} = \dots$, not just a vector.

State average $0.6 / 1$ (60%); 55% scored full marks. Mark distribution: 0: 45% · 1: 55%

Your mark: _____ / 1

b. Shortest distance from L_1 to A

- ☐ Vector from A to a point on L_1 : $(1, 3, -6)$
☐ Cross with direction: $|(15, 5, 5)| = 5\sqrt{11}$
☐ $\frac{5\sqrt{11}}{\sqrt{6}} = \frac{5\sqrt{66}}{6}$

Answer / notes: Use the given line, not your part (a) line.

State average $1.9 / 3$ (63%); 50% scored full marks. Mark distribution: 0: 26% · 1: 11% · 2: 13% · 3: 50%

Your mark: _____ / 3

c. Cartesian equation of plane ABC

- ☐ $\vec{AB} = (1, -3, -4)$, $\vec{AC} = (-1, 4, -8)$
☐ Normal $(40, 12, 1)$
☐ $40x + 12y + z = 19$

Answer / notes: Check B and C both give 19. CAS crossP avoids arithmetic slips.

State average $2.4 / 3$ (80%); 67% scored full marks. Mark distribution: 0: 15% · 1: 3% · 2: 15% · 3: 67%

Your mark: _____ / 3

d.i. Axis intercepts

- ☐ $(6, 0, 0)$, $(0, -4, 0)$, $(0, 0, 3)$

Answer / notes: Set two coordinates to zero each time.

State average $0.7 / 1$ (70%); 63% scored full marks. Mark distribution: 0: 37% · 1: 63%

Your mark: _____ / 1

d.ii. Area of the triangle

- ☐ Cross product of two side vectors: $(-12, 18, -24)$
☐ Area $= 3\sqrt{29}$

Answer / notes: Halve the cross product's magnitude; the triangle is neither isosceles nor right-angled.

State average $1.0 / 2$ (50%); 46% scored full marks. Mark distribution: 0: 44% · 1: 10% · 2: 46%

Your mark: _____ / 2

Question 4 (9 marks) · state average 6.0 / 9 (67%)

2024 VCAA Exam 2 · Section B · Question 6

a. Hypotheses

☐ $H_0 : \mu = 1000, H_1 : \mu < 1000$

Answer / notes: "Significantly less than" makes it one-tailed. Use μ .

State average 0.9 / 1 (90%); 91% scored full marks. Mark distribution: 0: 9% · 1: 91%

Your mark: _____ / 1

b.i. p-value to 3 d.p.

☐ 0.037

Answer / notes: $sd(\bar{X}) = \frac{4.2}{3} = 1.4$.

State average 0.9 / 1 (90%); 85% scored full marks. Mark distribution: 0: 15% · 1: 85%

Your mark: _____ / 1

b.ii. Should the machine be paused?

☐ Yes, since $p < 0.05$

Answer / notes: Answer the actual question (pause or not).

State average 0.8 / 1 (80%); 75% scored full marks. Mark distribution: 0: 25% · 1: 75%

Your mark: _____ / 1

c. Probability of a type II error

☐ Critical value $1000 - 1.6449 \times 1.4 \approx 997.70$

☐ $\Pr(\bar{X} > 997.70 \mid \mu = 997) \approx 0.31$

Answer / notes: Find the critical value under H_0 first, then use the true mean.

State average 1.0 / 2 (50%); 44% scored full marks. Mark distribution: 0: 45% · 1: 11% · 2: 44%

Your mark: _____ / 2

d. Values a and b

☐ $a \approx 996.7, b \approx 1003.3$

Answer / notes: $1000 \pm 2.3263 \times 1.4$.

State average 0.6 / 1 (60%); 58% scored full marks. Mark distribution: 0: 42% · 1: 58%

Your mark: _____ / 1

e. 95% confidence interval

☐ (1003.9, 1006.1)

Answer / notes: $1005 \pm 1.96 \times \frac{4}{\sqrt{50}}$.

State average 0.8 / 1 (80%); 82% scored full marks. Mark distribution: 0: 18% · 1: 82%

Your mark: _____ / 1

f. Expected number of the 40 intervals containing μ

☐ 38

Answer / notes: 0.95×40 , not 50.

State average 0.5 / 1 (50%); 51% scored full marks. Mark distribution: 0: 49% · 1: 51%

Your mark: _____ / 1

g. Minimum sample size

☐ 62

Answer / notes: $1.96 \times \frac{4}{\sqrt{n}} \leq 1 \Rightarrow n \geq 61.5$.

State average 0.5 / 1 (50%); 46% scored full marks. Mark distribution: 0: 54% · 1: 46%

Your mark: _____ / 1

Question 5 (10 marks) · state average 6.5 / 10 (65%)

2025 VCAA Exam 2 · Section B · Question 4

a. Starting point

☐ $(1, 0)$

Answer / notes: $\tilde{r}(0) = (5 - 4)\tilde{i}$. Give it as coordinates.

State average 0.9 / 1 (93%); 93.44% scored full marks. Mark distribution: 0: 6.56% · 1: 93.44%

Your mark: _____ / 1

b. Direction arrow at $(9, 0)$

☐ Arrow pointing upwards (anticlockwise)

Answer / notes: The particle is at $(9, 0)$ when $t = 2\pi$, with $\dot{y} = 15 > 0$.

State average 0.6 / 1 (62%); 62.76% scored full marks. Mark distribution: 0: 37.24% · 1: 62.76%

Your mark: _____ / 1

c. First return to the start

☐ $t = 4\pi$

Answer / notes: LCM of the periods 2π and $\frac{4\pi}{5}$. Give it exactly.

State average 0.5 / 1 (52%); 52.92% scored full marks. Mark distribution: 0: 47.08% · 1: 52.92%

Your mark: _____ / 1

d. Show the speed formula

☐ $\dot{x} = -5 \sin t + 10 \sin \frac{5t}{2}, \dot{y} = 5 \cos t - 10 \cos \frac{5t}{2}$

☐ $\text{Speed}^2 = 125 - 100 \left(\cos t \cos \frac{5t}{2} + \sin t \sin \frac{5t}{2} \right)$

☐ Compound-angle identity gives $125 - 100 \cos \frac{3t}{2}$

Answer / notes: Show the identity step: shortcuts lost marks.

State average 1.7 / 3 (58%); 35.56% scored full marks. Mark distribution: 0: 16.82% · 1: 27.94% · 2: 19.68% · 3: 35.56%

Your mark: _____ / 3

e. Maximum speed

☐ 15 m/s

Answer / notes: $\sqrt{125 + 100}$.

State average 0.7 / 1 (72%); 72.94% scored full marks. Mark distribution: 0: 27.06% · 1: 72.94%

Your mark: _____ / 1

f. Trace the path for $t \in [0, \pi]$

☐ From $(1, 0)$ to $(-5, -4)$ along the curve

Answer / notes: Stop exactly at the end point.

State average 0.7 / 1 (65%); 65.78% scored full marks. Mark distribution: 0: 34.22% · 1: 65.78%

Your mark: _____ / 1

g. Length of the traced path

☐ $\int_0^\pi \sqrt{125 - 100 \cos \frac{3t}{2}} dt$

☐ ≈ 36.6 m

Answer / notes: Write the definite integral, not just the number.

State average 1.4 / 2 (68%); 63.7% scored full marks. Mark distribution: 0: 27.2% · 1: 9.1% · 2: 63.7%

Your mark: _____ / 2

Question 6 (10 marks) · state average 5.9 / 10 (59%)

2025 VCAA Exam 2 · Section B · Question 5

a. Point of intersection

☐ $(-5, 2, 2)$

Answer / notes: Solve the three equations simultaneously.

State average 0.7 / 1 (73%); 73.51% scored full marks. Mark distribution: 0: 26.49% · 1: 73.51%

Your mark: _____ / 1

b.i. Direction of the line of intersection

- ☐ Cross product of the normals $(3, 6, 5) \times (1, 9, -3)$
☐ $-9\tilde{i} + 2\tilde{j} + 3\tilde{k}$ (or a multiple)

Answer / notes: $(3, 6, 5) \times (1, 9, -3) = (-63, 14, 21) = 7(-9, 2, 3)$.

State average 1.2 / 2 (62%); 51.97% scored full marks. Mark distribution: 0: 27.67% · 1: 20.36% · 2: 51.97%

Your mark: _____ / 2

b.ii. Parametric equations

- ☐ $x = -5 - 9\lambda, y = 2 + 2\lambda, z = 2 + 3\lambda$

Answer / notes: Parametric equations, not a Cartesian or vector form.

State average 0.5 / 1 (51%); 51.18% scored full marks. Mark distribution: 0: 48.82% · 1: 51.18%

Your mark: _____ / 1

c. Distance from $(1, 1, 2)$ to Π_3

- ☐ $\frac{|1+9-6-7|}{\sqrt{91}}$
☐ $\frac{3}{\sqrt{91}}$

Answer / notes: Show the method, not just the answer.

State average 1.4 / 2 (71%); 64.97% scored full marks. Mark distribution: 0: 22.49% · 1: 12.54% · 2: 64.97%

Your mark: _____ / 2

d.i. Show Π_1 is parallel to each plane in the family

- ☐ Normal $(6, 0, 27) = 3(2, 0, 9)$

Answer / notes: The normals are scalar multiples (factor 3).

State average 0.7 / 1 (67%); 67.92% scored full marks. Mark distribution: 0: 32.08% · 1: 67.92%

Your mark: _____ / 1

d.ii. Values of m

- ☐ Π_1 as $6x + 27z = 24$
☐ $\frac{|m-24|}{3\sqrt{85}} = \frac{23}{3\sqrt{85}}$
☐ $m = 1$ or 47

Answer / notes: Use the modulus to get both values.

State average 1.4 / 3 (45%); 33.79% scored full marks. Mark distribution: 0: 41.82% · 1: 13.61% · 2: 10.77% · 3: 33.79%

Your mark: _____ / 3

Your result

Total: _____ / 122 State average on these questions: **74 / 122 (61%)**

Topics to revise:

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